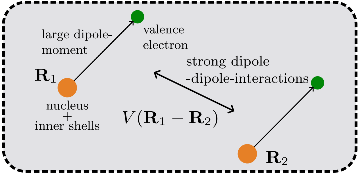
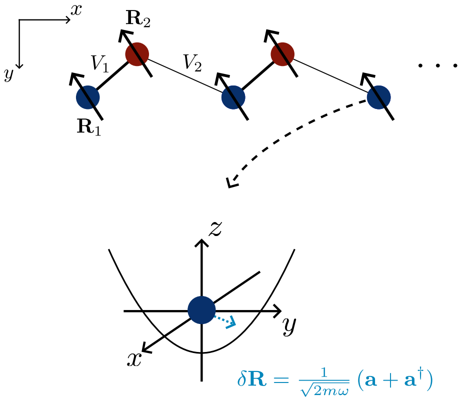
Rydberg atoms



preparing the atoms in the Rydberg states:



$$H_{\mathrm{eff}} = \sum_{i,j} U_{i,j} a_i^\dagger a_j$$

for $i \neq j$: $U_{ij} = \frac{1}{m\omega} \frac{\partial^2 V(\mathbf{R}_i - \mathbf{R}_j)}{\partial \delta \mathbf{R}_i \partial \delta \mathbf{R}_j}$

- **1D phonons:**
  - can be mapped to the extended ssh model
- **2D phonons:**
  - two independent nearest neighbor ssh models